(A) CONTINUOUS STATE SPACE

$$x(t) \rightarrow \boxed{\begin{aligned} h'(t) &= A\,h(t) + B\,x(t) \\ y(t) &= C\,h(t) + D\,x(t) \end{aligned}} \rightarrow y(t)$$

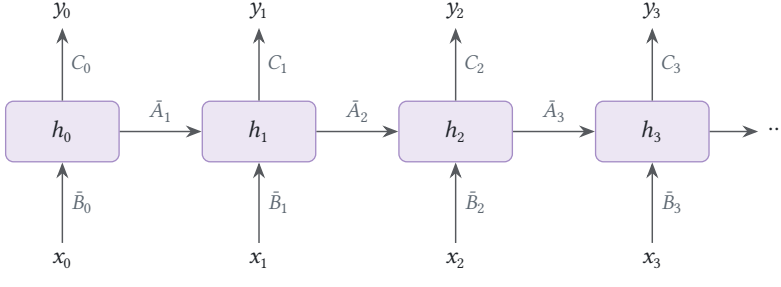
$h(t) \in \mathbb{R}^N$, A diagonal and negative

(B) DISCRETISATION, ZERO-ORDER HOLD

$$\begin{aligned} \bar{A} &= \exp(\Delta A) \\ \bar{B} &= (\Delta A)^{-1}(\exp(\Delta A) - I)\,\Delta B \approx \Delta B \end{aligned}$$

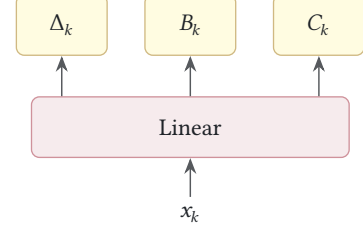
step size $\Delta > 0$; $\bar{A} \in (0, 1)$ per state

(C) RECURRENT VIEW, ONE STEP PER TOKEN



$h_k = \bar{A}_k h_{k-1} + \bar{B}_k x_k$, $y_k = C_k h_k + D x_k$. One pass over L tokens costs $O(L)$, and the state h is all the memory the model keeps.

(D) SELECTION (MAMBA)



$\Delta_k \to 0$: $\bar{A}_k \to 1$, $\bar{B}_k \to 0$, the state is held and x_k ignored.

Δ_k large: $\bar{A}_k \to 0$, the past is forgotten and the state rewritten from x_k.

Because Δ, B, C depend on the input, the model chooses per token what to keep. This is the knob CFFM-Net scales by sensor reliability.